郭子剑

联系方式	数据科学研究中心 浙江大学，中国	zijguo@zju.edu.cn `https://statweb.rutgers.edu/zijguo/`

教育背景	统计学博士, 宾夕法尼亚大学沃顿商学院 博士生导师: T. Tony Cai *"Statistical Inference For High-Dimensional Linear Models"*	2017
	数学理学学士, 香港中文大学 一级荣誉学士学位	2012

任职经历	求是讲席教授 数据科学研究中心 浙江大学	2025 年 8 月 - 至今
	副教授 (终身教职) 统计系 罗格斯大学——新泽西州立大学	2022 年 7 月 - 2025 年 7 月
	助理教授 统计系 罗格斯大学——新泽西州立大学	2017 年 9 月 - 2022 年 6 月

研究领域

优化与统计、多源学习、非标准推断、因果推断、高维推断，以及在遗传学和健康研究中的应用。

荣誉奖励

- ICSA 杰出青年学者奖 (Outstanding Young Researcher Award) 2023 年 7 月
- 伯努利学会 (Bernoulli Society) 新锐学者奖荣誉提名 2022 年 11 月
- ICSA 新锐学者奖 (New Researcher Award) 2019 年 12 月
- IMS 差旅奖 (Travel Award), JSM 2017 年 8 月
- 校长 Gutmann 领导力奖 (President Gutmann Leadership Award), 宾夕法尼亚大学 2017 年 4 月
- J. Parker Bursk 奖 2016 年 9 月

  由沃顿商学院颁发，以表彰在研究方面的卓越表现。
- 流行病学统计青年学者奖 (Statistics in Epidemiology Young Investigator Award), JSM 2013 年 8 月

  由美国统计学会 *(ASA)* 流行病学统计分会颁发。

学术访问

特邀研究员 (2018 年 11 月, 2023 年 9-10 月, 2024 年 4-5 月, 2025 年 6 月)
苏黎世联邦理工学院数学研究所 (邀请人: Peter Bühlmann)

特邀研究员 (2024 年 4-5 月, 2024 年 12 月, 2025 年 6 月)
法国国家信息与自动化研究所, 巴黎 (邀请人: Francis Bach)

特邀研究员 (2024 年 6 月)
牛津大学统计/经济系 (邀请人: Frank Windmeijer)

特邀研究员 (2024 年 4 月)
剑桥大学数学系 (邀请人: Richard Samworth)

特邀研究员 (2024 年 3 月)
特文特大学 (邀请人: Johannes Schmidt-Hieber)

特邀研究员 (2019 年 9 月, 2023 年 9 月, 2024 年 2-3 月)
哈佛大学陈曾熙公共卫生学院 (邀请人: Tianxi Cai)

特邀研究员 (2024 年 2 月)
耶鲁大学经济系 (邀请人: Xiaohong Chen)

特邀研究员 (2024 年 1-2 月)
卡内基梅隆大学统计与数据科学系 (邀请人: Larry Wasserman)

特邀研究员 (2017 年 8 月)
宾夕法尼亚大学佩雷尔曼医学院 (邀请人: Hongzhe Li)

代表性论文

* 表示作者按姓氏首字母排序；下划线 表示指导学生；# 表示同等贡献；✉ 表示共同通讯作者。

(R1) *Cai, T. T., and **Guo, Z.** (2017). Confidence intervals for high-dimensional linear regression: Minimax rates and adaptivity. *Annals of Statistics*, 45(2), 615-646.

(R2) **Guo, Z.**, Kang, H., Cai, T. T., and Small, D. S. (2018). Confidence Interval for Causal Effects with Invalid Instruments using Two-Stage Hard Thresholding. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80(4), 793-815.

(R3) **Guo, Z.**#, Ćevid, D.#, and Bühlmann, P. (2022). Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding. *Annals of Statistics*, 50(3), 1320-1347.

(R4) **Guo, Z.** (2023). Causal Inference with Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling. *Journal of the Royal Statistical*

Society: Series B (Statistical Methodology), 85(3), 959-985.

(R5) **Guo, Z.** (2024). Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies. *Journal of the American Statistical Association*, 119(547), 1968-1984.

(R6) Yao, M., Miller, G., Vardarajan, B., Baccarelli, A., **Guo, Z.** ✉, and Liu, Z. ✉. (2024+). Robust Mendelian Randomization Analysis by Automatically Selecting Valid Genetic Instruments with Applications to Identify Plasma Protein Biomarkers for Alzheimer's Disease. *Cell Genomics*, 2024 Dec 11;4(12):100700.

(R7) Wang, Z.#, Hu, Y.#, Bühlmann, P.✉, & **Guo, Z.**✉(2024). Causal Invariance Learning via Efficient Optimization of a Nonconvex Objective. *arXiv preprint arXiv:2412.11850.*

(R8) **Guo, Z.**, Wang, Z., Hu, Y., and Bach, F. (2025). Statistical Inference for Conditional Group Distributionally Robust Optimization With Cross-Entropy Loss. *arXiv preprint arXiv:2507.09905*

(R9) Zheng, M., Bonvini, M.✉, and **Guo, Z.**✉(2025). Perturbed Double Machine Learning: Nonstandard Inference Beyond the Parametric Length. *arXiv preprint arXiv:2511.01222*

预印本

* 表示作者按姓氏首字母排序；下划线 表示指导学生；# 表示同等贡献；✉ 表示共同通讯作者。

1. Heng, S.✉, Shen, Y., Guo, Z.✉(2026). Propensity Score Propagation: A General Framework for Design-Based Inference with Unknown Propensity Scores. *arXiv preprint arXiv:2601.13150*
2. Xiong, X., Guo, Z., Zhu, H., Hong, C., Smoller, J. W., Cai, T., Liu, M. (2026). Adversarial Drift-Aware Predictive Transfer: Toward Durable Clinical AI. *arXiv preprint arXiv:2601.11860*
3. Koo, T. and **Guo, Z.** (2025). Distributionally Robust Synthetic Control: Ensuring Robustness Against Highly Correlated Controls and Weight Shifts. *arXiv preprint arXiv:2511.02632*
4. Zheng, M., Bonvini, M.✉, and **Guo, Z.**✉(2025). Perturbed Double Machine Learning: Nonstandard Inference Beyond the Parametric Length. *arXiv preprint arXiv:2511.01222*
5. **Guo, Z.**, Wang, Z., Hu, Y., and Bach, F. (2025). Statistical Inference for Conditional Group Distributionally Robust Optimization With Cross-Entropy Loss. *arXiv preprint arXiv:2507.09905*
6. Wang, Z., Liu, M., Lei, J., Bach, F.✉, & **Guo, Z.**✉(2025). StablePCA: Learning Shared Representations across Multiple Sources via Minimax Optimization. *arXiv preprint arXiv:2505.00940.*

7. Scheidegger, C., **Guo, Z.**, & Bühlmann, P. (2025). Inference for Heterogeneous Treatment Effects with Efficient Instruments and Machine Learning. *arXiv preprint arXiv:2503.03530.*
8. Gu, Y., Fang, C., Xu, Y., **Guo, Z.**, & Fan, J. (2025). Fundamental Computational Limits in Pursuing Invariant Causal Prediction and Invariance-Guided Regularization. *arXiv preprint arXiv:2501.17354.*
9. Wang, Z.[#], Hu, Y.[#], Bühlmann, P., & **Guo, Z.** (2024). Causal Invariance Learning via Efficient Optimization of a Nonconvex Objective. *arXiv preprint arXiv:2412.11850.*
10. Rakshit, P., & **Guo, Z.** (2024). Statistical Inference in High-dimensional Poisson Regression with Applications to Mediation Analysis. *arXiv preprint arXiv:2410.20671.*
11. Wang, Z., Si, N., **Guo, Z.**, and Liu, M. (2024). Multi-source Stable Variable Importance Measure via Adversarial Machine Learning. *arXiv preprint arXiv:2409.07380.*
12. Zhan, K., Xiong, X., **Guo, Z.**, Cai, Tianxi, and Liu, M. (2024). Transfer Learning Targeting Mixed Population: A Distributional Robust Perspective. *arXiv preprint arXiv:2407.20073.*
13. *Fan, Q., **Guo, Z.**, Mei, Z., and Zhang, C. (2023). Uniform Inference for Nonlinear Endogenous Treatment Effects with High-Dimensional Covariates. *arXiv preprint arXiv:2310.08063.*
14. Xiong, X., **Guo, Z.**✉, and Cai, Tianxi ✉(2023). Distributionally Robust Transfer Learning. *arXiv preprint arXiv:2309.06534.*
15. Liu, Y., Liu, M., **Guo, Z.**, and Cai, Tianxi. (2023). Surrogate-Assisted Federated Learning of high dimensional Electronic Health Record Data. *arXiv preprint arXiv:2302.*
04970.
16. **Guo, Z.**, Zheng, M., and Bühlmann, P. (2022). Robustness Against Weak or Invalid Instruments: Exploring Nonlinear Treatment Models with Machine Learning. *arXiv preprint arXiv:2203.12808.*
17. ***Guo, Z.**, Yuan, W. and Zhang, C. (2019). Decorrelated Local Linear Estimator: Inference for Non-linear Effects in High-dimensional Additive Models. *arXiv preprint arXiv:1907.12732.* Reject and Resubmit at *Journal of Machine Learning Research.*

出版物

* 表示作者按姓氏首字母排序；下划线 表示指导学生；# 表示同等贡献；✉ 表示共同通讯作者。

1. Wang, Z., Bühlmann, P., and **Guo, Z.**(2025+). Distributionally Robust Learning for Multi-source Unsupervised Domain Adaptation. *Annals of Statistics*, to appear.

2. **Guo, Z.**, Li, X., Han, L., and Cai, Tianxi. (2025+). Robust Inference for Federated Meta-Learning. *Journal of the American Statistical Association*, to appear.
3. Chang, T. H., **Guo, Z.**, and Malinsky, D. (2025+). Post-selection inference for causal effects after causal discovery. *Biometrika*, to appear.
4. Scheidegger, C., **Guo, Z.**, and Bühlmann, P. (2024+). Spectral Deconfounding for High-Dimensional Sparse Additive Models. *ACM/IMS Journal of Data Science*, to appear.
5. Yao, M., Miller, G., Vardarajan, B., Baccarelli, A., **Guo, Z.** ✉, and Liu, Z ✉. (2024+). Robust Mendelian Randomization Analysis by Automatically Selecting Valid Genetic Instruments with Applications to Identify Plasma Protein Biomarkers for Alzheimer's Disease. *Cell Genomics*, to appear.
6. Lin, Y., **Guo, Z.**, Sun, B., and Lin, Z. (2024+). Testing High-Dimensional Mediation Effect with Arbitrary Exposure-Mediator Coefficients. *Test*, to appear.
7. Carl, D., Emmenegger, C., Bühlmann, B., **Guo, Z.** (2024+). TSCI: Two Stage Curvature Identification for Causal Inference with Invalid Instruments. *Journal of Statistical Software*, to appear.
8. *Fan, Q., **Guo, Z.**, and Mei, Z. (2024+). A Heteroskedasticity-Robust Overidentifying Restriction Test with High-Dimensional Covariates. *Journal of Business & Economic Statistics*, to appear.
9. **Guo, Z.** (2024). Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies. *Journal of the American Statistical Association*, 119(547), 1968-1984.
10. Kang, H., **Guo, Z.**, Liu, Z., and Small, D. (2024). Identification and Inference with Invalid Instruments. *Annual Review of Statistics and Its Application*, to appear.
11. Ma, R., **Guo, Z.**, Cai, T. T., and Li, H. (2024). Statistical Inference of Genetic Relatedness using High-Dimensional Logistic Regression. *Statistica Sinica*, 34 (2024): 1023-1043..
12. Rakshit, P., Wang, Z., Cai, T. T., and **Guo, Z.** (2024). `SIHR`: An R Package for Statistical Inference in High-dimensional Linear and Logistic Regression Models. *R Journal*, to appear.
13. *Cai, T. T., **Guo, Z.**, and Xia, Y. (2023). Statistical inference and large-scale multiple testing for high-dimensional regression models. *Test (with discussion)*, 32(4), 1135-1171.
14. **Guo, Z.** (2023). Causal Inference with Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 85(3), 959-985.

15. *Cai, T. T., **Guo, Z.**, and Ma, R. (2023). Statistical Inference for High-Dimensional Generalized Linear Models with Binary Outcomes. *Journal of the American Statistical Association*, 118 (542), 1319-1332.

16. Koo, T., Lee, Y., Small, D. S., and **Guo, Z.** (2023). `RobustIV` and `controlfunctionIV`: Robustiv and controlfunctioniv: Causal inference for linear and nonlinear models with invalid instrumental variables. *Observational Studies*, 9(4), 97-120.

17. Hou, J., **Guo, Z.**, and Cai, T. (2023). Surrogate assisted semi-supervised inference for high dimensional risk prediction. *Journal of Machine Learning Research*, 24(265), 1-58.

18. Wang, X., Zhou, H., $\cdots$, 4CE, Avillach, P.✉, **Guo, Z.**✉, and Cai, Tianxi✉. (2022) SurvMaximin: Robust Federated Approach to Transporting Survival Risk Prediction Models. *Journal of Biomedical Informatics*, 134 (2022): 104-176.

19. **Guo, Z.**, Ćevid, D., and Bühlmann, P. (2022). Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding. *Annals of Statistics*, 50 (3), 1320 - 1347.

20. ***Guo, Z.** and Zhang, C. (2022). Extreme Nonlinear Correlation for Multiple Random Variables and Stochastic Processes with Applications to Additive Models. *Stochastic Processes and their Applications*. 150, 1037-1058.

21. **Guo, Z.**, Renaux, C., Bühlmann, P., and Cai, T. T. (2021). Group Inference in High Dimensions with Applications to Hierarchical Testing. *Electronic Journal of Statistics*, 15(2), 6633-6676.

22. **Guo, Z.**, Rakshit, P., Herman, D., and Chen, J. (2021). Inference for Case Probability in High-dimensional Logistic Regression. *Journal of Machine Learning Research*, 22(254), 1-54.

23. *Cai, Tianxi, Cai, T. T., and **Guo, Z.** (2021). Optimal Statistical Inference for Individualized Treatment Effects in High-dimensional Models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 83(4), 669-719.

24. *Cai, T. T., and **Guo, Z.** (2020). Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(2), 391-419.

25. **Guo, Z.**, Wang, W., Cai, T. T., and Li, H. (2019). Optimal Estimation of Genetic Relatedness in High-dimensional Linear Models. *Journal of the American Statistical Association*, 114(525), 358-369.

26. **Guo, Z.**, Kang, H., Cai, T. T., and Small, D. S. (2018). Testing Endogeneity with High Dimensional Covariates. *The Journal of Econometrics*, 207(1), 175-187.

27. **Guo, Z.**, Kang, H., Cai, T. T., and Small, D. S. (2018). Confidence Interval

for Causal Effects with Invalid Instruments using Two-Stage Hard Thresholding. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80(4), 793-815.

28. *Cai, T. T., and **Guo, Z.** (2018). Accuracy assessment for high-dimensional linear regression. *Annals of Statistics*, 46(4), 1807-1836.
29. **Guo, Z.**, Small, D. S., Gansky, S. A., and Cheng, J. (2018). Mediation analysis for count and zero-inflated count data without sequential ignorability and its application in dental studies. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 67(2), 371-394.
30. Cheng, J., Cheng N. F., **Guo, Z.**, Gregorich, S., Amid I. I., and Gansky, S. A. (2018). Mediation analysis for count and zero-inflated count data. *Statistical Methods in Medical Research*, 27(9), 2756-2774.
31. *Cai, T. T., and **Guo, Z.** (2017). Confidence intervals for high-dimensional linear regression: Minimax rates and adaptivity. *Annals of Statistics*, 45(2), 615-646.
32. **Guo, Z.**, and Small, D. S. (2016). Control function instrumental variable estimation of nonlinear causal effect models. *Journal of Machine Learning Research*, 17(100), 1-35.
33. **Guo, Z.**, Cheng, J., Lorch, S. A., and Small, D. S. (2014). Using an instrumental variable to test for unmeasured confounding. *Statistics in Medicine*, 33(20), 3528-3546.
34. **Guo, Z.**, Kogan, R., Qiu, H., and Strichartz, R. S. (2014). Boundary value problems for a family of domains in the Sierpinski gasket. *Illinois Journal of Mathematics*, 58(2), 497-519.

软件

1. R 包 **SIHR**: statistical inference in high-dimensional regression.
   地址: `https://cran.r-project.org/web/packages/SIHR/index.html`
2. R 包 **RobustIV**: robust causal inference with possibly invalid instruments.
   地址: `https://cran.r-project.org/web/packages/RobustIV/index.html`
3. R 包 **TSCI**: two stage curvature identification with machine learning.
   地址: `https://cran.r-project.org/web/packages/TSCI/index.html`
4. R 包 **maczic**: mediation analysis for count and zero-inflated count data.
   地址: `https://cran.r-project.org/web/packages/maczic/index.html`
5. R 包 **MaximinInfer**: inference for maximin effects in high-dimensional settings.
   地址: `https://cran.r-project.org/web/packages/MaximinInfer/index.html`
6. R 包 **DLL**: inference for function derivative in high-dimensional additive models.
   地址: `https://cran.r-project.org/web/packages/DLL/index.html`

7. R 包 **DDL**: inference for regression parameters in high-dimensional models with hidden confounders.
   地址: `https://cran.r-project.org/web/packages/DDL/index.html`
8. R 包 **controlfunctionIV**: control function method with possibly invalid instruments.
   地址: `https://cran.r-project.org/web/packages/controlfunctionIV/index.html`

R 代码均可在 `https://github.com/zijguo` 找到.

过往科研项目

1. 美国国立卫生研究院
   R01AG086379 "Robust Mendelian Randomization Framework with Multi-Omics Data for Alzheimer's Disease and Related Dementias"
   - 职责: 共同项目负责人 (Co-PI) (负责人: Dr. Zhonghua Liu)
2. 美国国立卫生研究院
   R01LM013614 "Semi-supervised Approaches to Denoising Electronic Health Records Data for Risk Prediction"
   - 职责: 共同项目负责人 (Co-PI) (负责人: Dr. Tianxi Cai)
3. 美国国立卫生研究院
   R01GM140463 "Predictive Modeling with High-Dimensional Incomplete Data."
   - 职责: 项目负责人 (PI) (共同负责人: Dr. Jinbo Chen)
4. 美国国家科学基金会
   DMS 2015373 "Repro Sampling Method: A Transformative Artificial-Sample-Based Inferential Framework with Applications to Discrete Parameter, High-Dimensional Data, and Rare Events Inferences."
   - 职责: 共同项目负责人 (Co-PI) (负责人: Dr. Min-ge Xie)
5. 美国国家科学基金会
   DMS 1811857 "Inference in High-Dimensional Linear Models: Methods, Theory and Applications."
   - 职责: 项目负责人 (PI)
6. 美国国立卫生研究院
   R56-HL-138306-01 "Statistics Methods for Analyzing Electronic Health Record Data."
   - 职责: 共同研究员 (Co-Investigator) (负责人: Dr. Jinbo Chen)
7. 宾夕法尼亚大学医学院
   "Statistics Methods for Analyzing Electronic Health Record Data."

- 职责: 高级研究员 (Senior Investigator) (负责人: Dr. Jinbo Chen)

授课经历

授课讲师

- **罗格斯大学 (博士生课程)**

STAT 593: Theory of Statistics	2024 年秋季
STAT 593: Theory of Statistics	2023 年秋季
STAT 594: Advanced Modern Statistical Inference II	2019 年春季
授课评分 *(Instructor Rating):* 4.82 / 5.0	

- **罗格斯大学 (研究生课程)**

FSRM 588: Financial Data Mining	2025 年春季
FSRM 588: Financial Data Mining	2021 年秋季
授课评分 *(Instructor Rating):* 4.75 / 5.0	
FSRM 588: Financial Data Mining (线上课程)	2021 年春季
FSRM 588: Financial Data Mining	2020 年春季
授课评分 *(Instructor Rating):* 4.50 / 5.0	
FSRM 588: Financial Data Mining	2019 年秋季
授课评分 *(Instructor Rating):* 4.75 / 5.0	
FSRM 588: Financial Data Mining	2018 年秋季
授课评分 *(Instructor Rating):* 4.71 / 5.0	
FSRM 588: Financial Data Mining	2017 年秋季
授课评分 *(Instructor Rating):* 4.82 / 5.0	

- **罗格斯大学 (本科生课程)**

STAT 384: Intermediate Statistical Analysis (线上课程)	2021 年春季
STAT 484: Basic Applied Statistics (线上课程)	

- **宾夕法尼亚大学沃顿商学院**

STAT 111 : Introductory Statistics	2016 年夏季

习题课讲师 2014 年秋季

宾夕法尼亚大学沃顿商学院

STAT 111: Introductory Statistics

助教

宾夕法尼亚大学沃顿商学院

STAT 102: Business Statistics	2017 年春季
STAT 970: Mathematical Statistics	2016 年秋季

STAT 622: Statistical Modeling 2016 年春季

STAT 550: Mathematical Statistics 2015 年秋季

学生指导

博士生导师:

1. Prabrisha Rakshit (2023 年毕业；Assistant Professor in the Department of Operations Management, Quantitative Methods, and Information Systems at the Indian Institute of Management Udaipur);
2. Taehyeon Koo (2025 年毕业；与 Dr. Nicole Pashley 联合指导；首职: Postdoctoral fellow at Columbia University);
3. Zhenyu Wang (罗格斯大学, 预计 2027 年毕业);
4. Mengchu Zheng (罗格斯大学, 预计 2028 年毕业);
5. Chao Qin (浙江大学)
6. Yuebei Shi (浙江大学)
7. Chen Ling (浙江大学)
8. Jiahang Shao (浙江大学)
9. Wenyuan Shu (浙江大学)

博士后合作导师:

1. Yichuan Bai (Ph.D. at Iowa State)

学术服务

1. 副主编 (Associate Editor)
   - *Journal of the American Statistical Association*, 2023- 至今
2. 因果推断阅读小组组织者 (Organizer of Causal Inference Reading Group)
   (与 Nicole Pashley 和 Tirthankar Dasgupta 共同组织), 罗格斯大学统计系
3. 系年度研讨会主席 (Department Retreat Chair, 2019-2020), 罗格斯大学统计系
4. 系学术报告会主席 (Department Seminar Chair, 2018-2019), 罗格斯大学统计系
5. 其他罗格斯大学委员会服务:
   - 系网站委员会 (Department website committee, 2020-2023)
   - 系年度研讨会委员会 (Department retreat committee, 2017-2018)
   - FSRM/MSDS 项目委员会 (FSRM/MSDS committee, 2017-2023)
   - 博士生考试委员会 (Ph.D. Exam committee, 2018-2020)
   - 研究生课程委员会 (Graduate Curriculum committee, 2019-2021)
6. 2019 年罗格斯统计研讨会组织委员会 (Organizing Committee for 2019 Rutgers Statistics Symposium)

7. ICSA 2019 第 11 届国际会议程序委员会 (Program Committee for ICSA 2019 11th International Conference)
8. 2018 年 ICSA 应用研讨会地方组织委员会 (Local Organizing Committee for 2018 ICSA Applied Symposium)
9. 期刊审稿人 (Reviewer for the following journals): *Annals of Statistics*, *Journal of the American Statistical Association*, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *Biometrika*, *Journal of Machine Learning*, *Journal of Econometrics*, *Biometrics*,*Statistica Sinica*, *IEEE International Symposium on Information Theory*, *Journal of Applied Statistics*, *COLT*.

短期课程

1. *"Nonregular Inference"*, 耶鲁大学经济系, 2024 年 2 月
2. *"Nonregular Inference"*, CELEHS Distinguished Lecture Series, 哈佛大学生物统计系, 2024 年 2 月
3. *"Robust Deconfounding"*, 哈佛大学生物统计系, 2024 年 2 月

学术报告

1. 系学术报告, 牛津大学经济系, 英国, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, 2024 年 6 月
2. 系学术报告, 牛津大学统计系, 英国, *"Robust Instrumental Variable Analysis."*, 2024 年 6 月
3. 系学术报告, 日内瓦大学, 瑞士, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, 2024 年 5 月
4. 系学术报告, 洛桑联邦理工学院, 瑞士, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, 2024 年 5 月
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6. 系学术报告, 法国国家信息与自动化研究所, 法国, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, 2024 年 4 月
7. 学术报告, 阿姆斯特丹机器学习实验室, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, 2024 年 3 月
8. 系学术报告, 耶鲁大学经济系, 美国, *"Robust Instrumental Variable Analysis."*, 2024 年 2 月
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19. 特邀演讲 (线上), EcoSta 2023, 日本东京, *"Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables."*, 2023 年 8 月

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28. 特邀报告 (线上), CMStatistics 2022, 英国伦敦, *“Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding.”*, 2022 年 12 月
29. Levin 讲座, 哥伦比亚大学生物统计系, 美国, *“Multi-data Replicability via the Maximin Criterion: High-Dimensional Inference for Group Distributionally Robust Models.”*, 2022 年 9 月
30. 实验室研讨会 (线上, 由 Tianxi Cai 教授主持), 哈佛大学陈曾熙公共卫生学院, 美国, *“Robust Inference for Federated Meta-Learning.”*, 2022 年 9 月
31. RAND 中心因果推断研讨会 (线上), *“Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables.”*, 2022 年 8 月
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35. 因果推断中心研讨会 (线上), 宾夕法尼亚大学, 美国, *“Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables.”*, 2022 年 3 月
36. 系学术报告 (线上), 威斯康星大学麦迪逊分校生物统计与医学信息学系, 美国, *“Decorrelated Local Linear Estimator: Inference for Non-linear Effects in High-dimensional Additive Models.”*, 2022 年 2 月
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38. 特邀报告人 (线上), CMStatistics 2021, *“Inference for Case Probability in High-dimensional Logistic Regression.”*, 2021 年 12 月

39. 特邀报告人 (线上), JSM 2021, 美国西雅图, *"Inference for High-dimensional Maximin Effects in Heterogeneous Regression Models Using a Sampling Approach."*, 2021 年 8 月

40. 特邀报告人 (线上), 第一届统计及相关领域国际会议, 卢森堡, *"Inference for High-dimensional Maximin Effects in Heterogeneous Regression Models Using a Sampling Approach."*, 2021 年 7 月

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49. 特邀报告 (线上), JSM 2020, 美国费城, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding and Measurement Errors."*, 2020 年 8 月

50. 特邀参会者 (线上), *"Mathematical and Statistical Challenges in Uncertainty Quantification."*, 英国剑桥, 2020 年 7 月

51. 系学术报告 (线上), 加州大学戴维斯分校统计系, 美国, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding and Measurement Errors."*, 2020 年 5 月

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53. 生物统计阅读小组报告 (线上, 由 Jinbo Chen 教授主持), 宾夕法尼亚大学, 美国, *"Group Inference in High Dimensions with Applications to Hierarchical Testing."*, 2020 年 5 月
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55. 特邀报告, 第 11 届 ICSA 国际会议, 中国杭州, *"Group Inference in High Dimensions with Applications to Hierarchical Testing."*, 2019 年 12 月
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77. 特邀报告, 英国剑桥, *"Statistical Foundations of Uncertainty Quantification for Inverse Problems."*, 2017 年 6 月

78. 研讨会, 宾夕法尼亚大学大数据统计方法中心, 美国, *"Inference with High-dimensional Covariates and Possibly Invalid Instruments."*, 2017 年 4 月

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其他报告

1. 专题投稿报告, Joint Statistical Meetings, 美国巴尔的摩, *"Optimal Estimation of Co-Heritability in High-Dimensional Linear Models"*, 2017 年 8 月
2. 投稿报告, Joint Statistical Meetings, 美国芝加哥, *"Accuracy Assessment for High-dimensional Linear Regression"*, 2016 年 8 月
3. 投稿报告, Eastern North American Region (ENAR), 美国奥斯汀, *"Confidence Intervals for High-Dimensional Linear Regression: Minimax Rates and Adaptivity"*, 2016 年 3 月
4. 海报展示, John W. Tukey Conference, 普林斯顿大学, *"Confidence Intervals for High-Dimensional Linear Regression: Minimax Rates and Adaptivity"*, 2015 年 9 月
5. 投稿报告, Joint Statistical Meetings, 美国西雅图, *"Distance Matrix Estimation from Noisy Observation of Low Rank Position Matrix"*, 2015 年 8 月
6. 投稿报告, Joint Statistical Meetings, 美国波士顿, *"Instrumental Variable Approach for Mediation Analysis of Count Model"*, 2014 年 8 月
7. 专题投稿报告, Joint Statistical Meetings, 加拿大蒙特利尔, *" Instrumental Variable Approach for Mediation Analysis of Zero-Inflated Count Model"*, 2013 年 8 月
8. 海报展示, Atlantic Causal Inference Conference (ACIC), 哈佛大学, *"Control function Instrumental Variable Estimation of Nonlinear Causal Effect Models"*, 2013 年 5 月

学术会员

- 数理统计学会 (Institute of Mathematical Statistics)
- 美国统计学会 (American Statistical Association)
- 英国皇家统计学会 (Royal Statistical Society)
- 伯努利数理统计与概率学会 (Bernoulli Society for Mathematical Statistics and Probability)
- 泛华统计协会 (International Chinese Statistical Association)
- 计量经济学会 (The Econometric Society)